

UWB Catheter Position Mapping with SAR-Bounded Accuracy

Executive Technology Brief

Domain: Radio Frequency (UWB) Localization + Tissue Propagation

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Catheter position verification requires CT/MRI (expensive, ionizing for CT)

Mechanism: UWB transmitter in catheter; external receiver array localizes via TOA with SAR-bounded accuracy (R1 fix)

Architecture: Implantable UWB transmitter in shunt catheter; external receiver array localizes catheter position via time-of-arrival (TOA) ranging. R1 fix: original P-21 assumed sub-mm accuracy without SAR analysis; R1 redesign bounds accuracy by SAR limit and tissue propagation physics.

Governing Model: UWB time-of-arrival ranging through tissue. Localization accuracy bounded by signal bandwidth, SNR (limited by SAR-constrained transmit power and tissue attenuation), and array geometry (GDOP).

Key Equations:

$d = c * dt$ [P-21-R1 UWB TOA from implant to external receiver array]

$SAR = (\sigma * E^2) / (2 * \rho)$ [P-21-R1 SAR constraint on UWB implant transmit power]

$\sigma_{TOA} \geq c / (2 * \pi * \beta * \sqrt{2 * SNR})$ [Cramer-Rao lower bound]

$\sigma_{pos} = GDOP * \sigma_{TOA}$ [geometric dilution of precision]

$PL_{tissue}(d, f) = \alpha(f) * d$ [frequency-dependent path loss]

Kill Condition: SAR limit (1.6 W/kg) constrains transmit power below useful SNR at >5cm tissue depth

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.